

Phase 2 — Master System Specification & Strategic Blueprint

Cloud E-Archive | Hub-and-Spoke Dispersal | Barcode PWA | Executive Dashboard

Built on Phase 1: 125,042 patient records migrated | Live CRM dashboard | Automated birthday messaging

EXECUTIVE SUMMARY

Phase 2 transforms clinical records operations at the University of Ghana Dental School. Building on Phase 1 — which migrated 125,042 patient records to the cloud and deployed a live CRM dashboard — Phase 2 permanently replaces the 15-year-old legacy desktop eArchive software with a cloud-native system that gives every stakeholder in the hospital real-time visibility of every patient folder, from the moment it is pulled from the shelf to the moment it is returned.

The system is built around a single core principle: **one scan, one destination selection, and everything updates everywhere simultaneously — the clinic queue, the records room, and the executive dashboard.**

Out of Scope Notice: X-Ray patients (who do not have folders) are captured as a future user requirement and are out of scope for Phase 2. They will be addressed in a subsequent phase.

1. WHY PHASE 2: THE LEGACY SYSTEM PROBLEM

For over fifteen years, all 125,042 patient folder histories have lived on a single desktop PC running Visual Basic eArchive software connected to a local MySQL 5.0 database on port 3307. This creates four critical operational risks:

Risk	Impact
Single Point of Failure	One hard drive crash or malware attack instantly paralyses records retrieval across all 13 departments.
Zero Clinical Custody	Once a folder leaves the records room, no one knows where it is. Staff waste hours searching clinic by clinic.
No Patient Journey History	The legacy system stores registration numbers only. It cannot record when a patient arrived, which clinic treated them, or how long they waited.
Management Blindspot	The Dean and leadership have no real-time counters of daily patient visits, forcing staffing decisions based on guesswork.

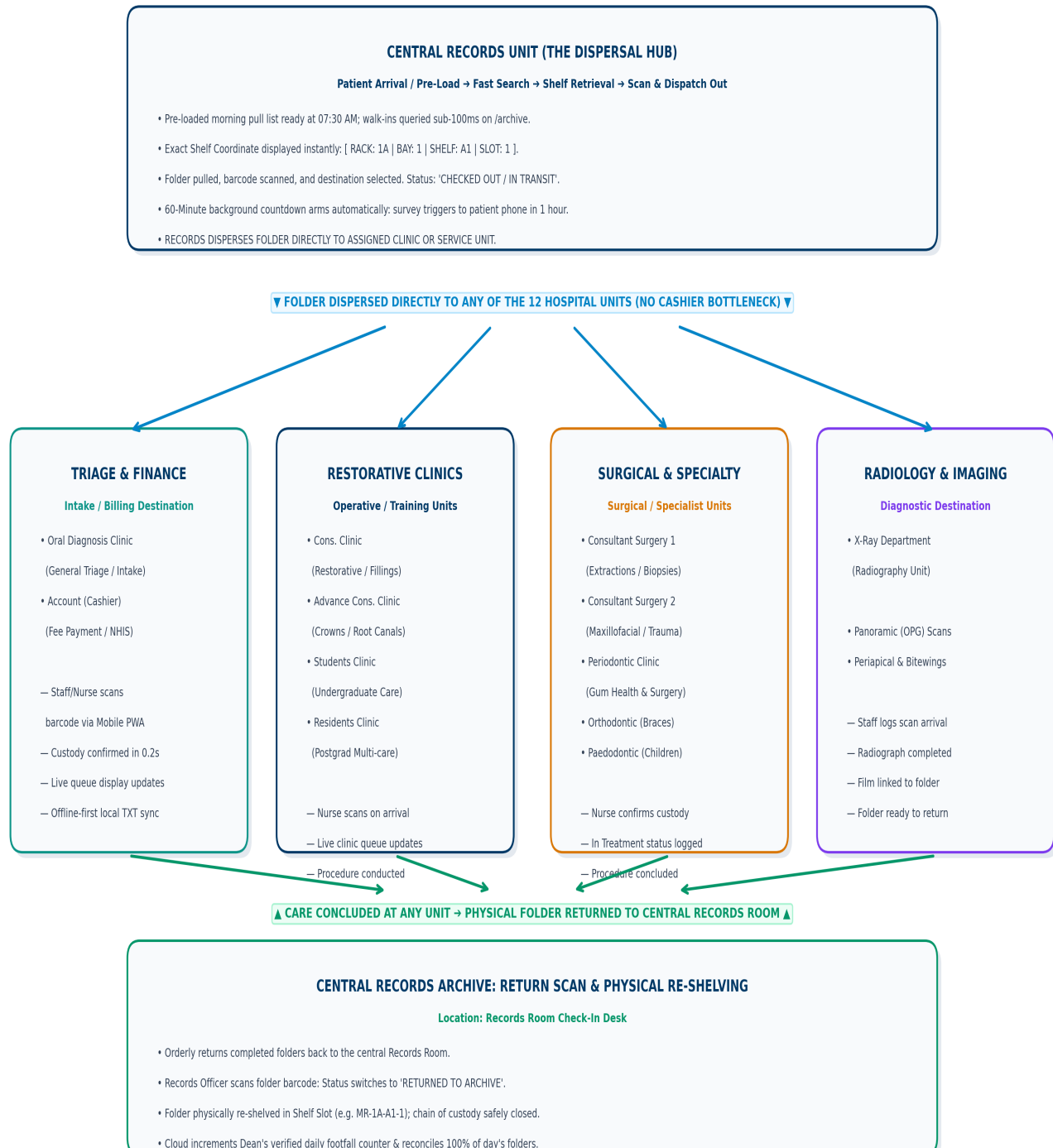
2. THE 13 UGDS CLINICAL UNITS

The Records Unit acts as the Central Archival and Dispersal Hub. All 12 clinical and service departments are spokes that receive folders directly from Records and return them directly to Records after treatment:

No.	Unit	Function	Role in Dispersal Model
1	Records Unit (Hub)	Central Archival & Dispersal	Pulls folders; scans out; selects destination; audits all returns.
2	Account (Cashier)	Billing & Financial Clearance	Independent service point; does not block clinical flow.
3	Oral Diagnosis Clinic	Primary Intake & Triage	Receives and routes new patients to specialist care.
4	Cons. Clinic	Conservative & Operative Dentistry	Receives folders for restorations and routine care.
5	Students Clinic	Undergraduate Training	Receives folders for supervised student clinical work.
6	Advance Cons. Clinic	Endodontics & Prosthodontics	Receives folders for root canals, crowns, and bridges.
7	Residents Clinic	Postgraduate Specialized Care	Receives folders for advanced referral cases.
8	Periodontic Clinic	Gum Health & Periodontology	Receives folders for scaling, surgery, and bone-loss care.
9	Paedodontic Clinic	Child & Adolescent Dentistry	Receives folders for child dentistry and preventive care.
10	Orthodontic Clinic	Malocclusion & Alignment	Receives folders for brace adjustments and follow-ups.
11	Consultant Surgery 1	Oral & Maxillofacial Surgery	Receives folders for surgical extractions and biopsies.
12	Consultant Surgery 2	Oral & Maxillofacial Surgery	Receives folders for facial trauma and surgical consultations.
13	X-Ray (Radiology)	Dental Imaging	Receives folders for periapical, bitewing, and OPG imaging.

The diagram below illustrates the direct Hub-and-Spoke Dispersal architecture: Records pulls from shelf coordinates, scans out to any of the 12 destination units, arming the 60-minute patient survey, and converges back to Records for re-shelving:

Hub & Spoke Model: Central Records Disperses Directly to Any Clinical / Service Unit → Re-Shelving



3. THE SEVEN SYSTEM COMPONENTS

3.1 Cloud E-Archive — Records Desk Interface

The records officer's primary interface. Replaces the legacy desktop software entirely.

Feature	Detail
Patient Search	Sub-100ms search across all 125,042 records by folder number or patient name.
Shelf Coordinate Locator	Instantly displays exact physical location: RACK BAY SHELF SLOT so officer walks directly to the right shelf.
Scan & Dispatch	Officer scans the folder barcode, selects destination clinic from dropdown. One action triggers everything.
Status Tracking	Folder status updates hospital-wide: PULLED / IN TRANSIT / IN CUSTODY / RETURNED TO ARCHIVE.

3.2 Morning Pre-Load & Pull List — Appointment Scheduling

Every patient at UGDS falls into one of two categories:

Patient Type	How They Are Handled
Appointment Patient	Has a pre-booked session with a specific doctor. Folder is pre-loaded into the system the night before or early morning so Records can prepare the morning pull list in advance.
Walk-in Patient	Arrives without an appointment. Folder is pulled and dispatched on arrival using the same scan-and-select flow.

Even appointment patients must go through the scan-and-select step at Records before their folder moves. The pre-load simply means Records knows in advance which folders to prepare.

3.3 Hub-and-Spoke Dispersal — The Core Workflow

The Records officer scans the barcode and selects the destination clinic from a dropdown. That single action triggers four simultaneous events:

What Happens	Where It Occurs
Folder status changes to IN TRANSIT	Records desk and all system views
Patient is added to that clinic's daily queue	Clinic PC / laptop display
Executive dashboard updates workload count	Dean's dashboard in real time
60-minute SMS survey countdown begins	Cloud engine (background dispatch)

3.4 Clinic Queue Display — On Each Clinic PC / Laptop

Each of the 13 clinics has its own display on their PC or laptop showing their patient queue for the day:

Queue Status	What the Clinic Sees
Expected	Patients dispatched from Records, folder in transit to this clinic.
Arrived	Folder scanned in by clinic nurse via the PWA. Patient confirmed in custody.
In Treatment	Patient currently being attended to by the doctor.
Completed	Treatment done. Folder awaiting return to Records.
Did Not Attend	Patient was expected but folder was never scanned in at the clinic.

3.5 Offline-First Local Sync — Clinic Data Resilience

The clinic queue and all scan activity at each clinic operates offline-first. This means:

Scenario	What Happens
Internet is available	All scans and queue updates sync to the central Neon PostgreSQL database in real time.
Internet drops	All activity continues normally. Scans and updates are saved immediately to a local TXT file on the clinic laptop. No data is lost.
Internet is restored	The system automatically detects the connection and syncs all locally saved records to the central database. No manual action required.

3.6 Staff Barcode PWA — Clinic Nurse & Assistant Interface

A mobile-first Progressive Web App that runs on any clinic smartphone, tablet, or PC browser. No app store installation required.

Feature	Detail
Authentication	Nurse or assistant signs in with their staff PIN and selects their active clinic unit.
Folder Arrival Scan	Live camera viewfinder with a single Scan Arriving Folder button. Nurse scans barcode upon arrival.
Custody Confirmation	Screen flashes green: Folder K/13 5622 Confirmed — In Custody at Cons. Clinic at 09:42 AM.
Instant Sync	Custody transfers in 0.2 seconds. Records room and executive dashboard update immediately.

3.7 Executive Dashboard — Dean & Management View

Real-time operational intelligence for the Dean and hospital leadership. No manual compilation required.

Metric	What It Shows
Live Daily Footfall	Verified total patients who visited today, this week, and this month across all 13 departments.
Departmental Workload	Live count of how many folders are currently held in each of the 13 clinical units.
Patient Journey Timeline	Full day timeline for any individual patient: 08:15 Records > Dispatched > 08:40 Confirmed > 10:15 Archived.
Patient Satisfaction	Live satisfaction scores from automated SMS surveys, integrated in real time.

4. A TYPICAL DAY: END-TO-END WORKFLOW

Time	Who	Action
07:30 AM	Records Officers	Open /archive. Pre-loaded appointment list is ready. Officers pull folders for scheduled patients and begin scanning and dispatching to clinics.
08:00 AM+	Walk-in Patients	Arrive at reception. Records officer searches folder, pulls from shelf coordinate, scans out, selects clinic. Patient added to clinic queue instantly.
08:15 AM	Clinic Nurse	Opens Staff PWA. Sees incoming folders on clinic queue. Scans each folder barcode as it arrives. Custody confirmed in 0.2s.
08:30 – 14:00	Doctors & Staff	Attend to patients. Clinic queue tracks who has been seen and who is still waiting in real time.
60 Min Post	Cloud Engine	Automated SMS dispatched to patient: personalized feedback survey link.
14:30 – 16:30	Records Officers	Treated folders returned from clinics. Officer scans each one back in. Status returns to ARCHIVED. Folder returned to exact shelf slot.
16:35 PM	Dean / Leadership	Reviews executive dashboard: total patients seen, clinic workload distribution, folder accountability, live satisfaction scores.

5. PATIENT TYPES & SYSTEM SCOPE

Patient Type	Folder?	Pre-Load?	Scan Flow?	Phase
Walk-in	Yes	No — pulled on arrival	Full scan-and-select	Phase 2 (Active)
Appointment	Yes	Yes — pre-loaded night before/morning	Full scan-and-select (mandatory)	Phase 2 (Active)
X-Ray Only	No	N/A	Future requirement — out of scope	Deferred to Future Phase

6. CLOUD INFRASTRUCTURE

Phase 2 moves from the single-PC legacy environment to a dedicated annual enterprise cloud package on Vercel and Neon PostgreSQL:

Component	Provider & Tier	Purpose	Key Safeguards
Frontend & API	Vercel Production Pro	Hosts E-Archive desk, clinic queues, and serverless API.	99.99% SLA uptime, regional CDN, automated SSL, custom UGDS domain.
Central Database	Neon PostgreSQL Production	Stores 125,042 patient records, folder movement logs, and staff data.	Automated point-in-time recovery, daily backups, connection pooling for all 13 clinics.
Staff PWA	Vercel PWA / Edge Cache	Delivers the mobile barcode scanner to all clinic devices.	Instant mobile loading, offline caching, no app store installation required.

7. IMPLEMENTATION: 4-WEEK ROLLOUT

Sprint	Focus	Key Deliverables
Week 1	Foundation & Cloud Setup	Configure on-site daemon at Records PC. Activate Vercel Pro and Neon production packages. Ingest 134 staff profiles. Set up appointment pre-load system.
Week 2	Cloud E-Archive Deployment	Deploy /archive module with sub-100ms search and shelf coordinate locator. Decommission legacy MySQL 5.0 desktop software.
Week 3	Clinic Queue & PWA Launch	Launch clinic queue displays on all 13 clinic PCs. Deploy Staff Barcode PWA. Activate offline-first local TXT sync. Roll out across Oral Diagnosis, Cons. Clinic, and Surgery.
Week 4	Executive Dashboard & Handover	Connect live footfall counters to Phase 1 dashboard. Conduct nursing staff training. Executive leadership review and sign-off.

8. FUTURE REQUIREMENTS (OUT OF SCOPE — PHASE 2)

Requirement	Description	Reason Deferred
X-Ray Patient Tracking	X-Ray patients do not have physical folders. A lightweight check-in system is needed to log their visits for daily footfall counts without triggering standard folder scan-and-dispatch.	Different workflow from folder-based patients. Requires separate design and will be scoped as a standalone feature in the next phase.

University of Ghana Dental School — Phase 2 Master System Specification | 2026

Cloud E-Archive | Hub-and-Spoke Dispersal | Barcode PWA | Executive Dashboard